SENTINEL: AN INTEGRATED SECURITY OPERATIONS MANAGEMENT SYSTEM FOR DAVAO SECURITY & INVESTIGATION AGENCY, INC.

A Capstone Project
Proposal Presented to the Faculty of the
Information and Communications Technology Program
STI College Tagum

In Partial Fulfillment
of the Requirements for the Degree Bachelor
of Science in Information Systems

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **SENTINEL: An Integrated Security Operations Management System**

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APPROVAL SHEET

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INTRODUCTION

Project Context

The private security global industry is transitioning toward automated, high-integrity digital ecosystems to address the discovery lag inherent in human-led monitoring. Global research in workforce capacity indicates that manual, reactive responses to personnel gaps significantly increase vacancy time, which remains a primary driver of security breaches in critical infrastructure (Shiyanbola et al., 2023). This movement is supported by modern performance management theories which emphasize that transparent, data-driven tracking is essential to maintaining accountability and preventing social loafing, a phenomenon where individual reliability decreases in the absence of structured, real-time monitoring (Aguinis, 2022).

The necessity for these technological advancements is echoed in recent Philippine legislation through the Private Security Services Industry Act, also known as Republic Act No. 11917 (2022). This law introduced a regime of professionalized accountability where agencies face strict liability for administrative negligence. Under the law's Implementing Rules and Regulations, agencies can be fined up to P5,000,000 or face license revocation for deploying personnel with expired licenses or unauthorized firearms (Jur.ph, 2025). Furthermore, national studies on the effectiveness of Philippine security guards identify that absenteeism and the abandonment of posts are most prevalent in agencies that lack advanced patrol monitoring and digital reporting systems (Abad, 2025).

In the specific context of the Davao Region, the Davao Security and Investigation Agency, Inc. operates in a landscape that has prioritized a culture of security to sustain its position as a leading safe haven in Southeast Asia (PIA, 2025). According to the Davao Regional Development Plan 2023 to 2028, the region's rapid economic growth has increased the demand for advanced technology and innovation to ensure peace and security in high-traffic logistics and commercial hubs like Tagum City (NEDA XI, 2024). Recent local research conducted in the Davao Region specifically highlights that compensation, punctuality, and alertness are the primary determinants of guard performance, yet many regional agencies still struggle with inattentiveness and abandonment of posts due to inconsistent manual oversight (Ondos and Origines, 2025). For DASIA Tagum, this represents a critical vulnerability where site vacancies may go undetected for hours, directly compromising the safety of local clients.

SENTINEL is an integrated digital ecosystem designed to resolve these cumulative vulnerabilities by centralizing personnel profiles, firearm telemetry, and shift logs into a high-performance single source of truth. The system works by embedding real-time compliance checks into the daily operational workflow. Before a guard can be deployed, the system automatically verifies their license validity and firearm authorization against central records. To manage high-concurrency requests from hundreds of guards, SENTINEL utilizes the Rust programming language and the Axum framework, providing native memory safety and asynchronous performance that prevents the data races typical of legacy systems (Scofield, 2025). When an operational gap such as a missed check-in is detected, the system immediately flags the vacancy and identifies the nearest qualified replacement, ensuring that the agency moves from a reactive stance to a proactive, legally defensible model of security management.

Purpose and Description

The primary purpose of this capstone project is to deliver a fully functional, production-ready Integrated Security Operations Management Platform that consolidates guard personnel management, equipment allocation, vehicle operations, and access control into a unified system for Davao Security & Investigation Agency, Inc.

SENTINEL is a web-based system that integrates operational data and workflows using a Rust/Axum backend API, a PostgreSQL relational database, and a React/TypeScript frontend deployed through Docker. The system is designed as an integrated operations platform that aligns with ERP and WFM principles by centralizing data, enforcing consistent workflows, and providing real-time operational visibility.

The platform implements four core modules. Guard Management covers personnel profiles, scheduling, attendance tracking, performance analytics, and replacement coordination to ensure uninterrupted site coverage. Equipment Management maintains firearm inventory, allocation workflows, permit records, and maintenance history to support regulatory compliance and accountability. Vehicle Operations manages armored vehicle assets, driver assignments, and trip tracking to improve deployment oversight. Access Control enforces role-based permissions, authentication, and audit logging to ensure that only authorized roles can access sensitive operations and data.

Objectives

General Objectives

To design, implement, and validate SENTINEL as a multi-platform Security Operations Center (SOC) platform that unifies personnel deployment, resource control, incident response, and command-level monitoring across Web, Desktop (Tauri), and Mobile (Capacitor Android) runtime targets.

Specific Objectives

1. Implement a complete account and identity lifecycle module for all users.
    a. Provide user registration, email verification, resend verification, and secure login.
    b. Implement forgot password, reset code verification, password reset, and guard approval gating for controlled account activation.
    c. Support profile management, profile photo update/delete, and secure session continuity through refresh-token rotation and logout revocation flows.
2. Implement centralized personnel administration and approval workflows.
    a. Manage users by role: superadmin, administrator, supervisor, and guard.
    b. Provide pending guard approval queues and approval status updates.
    c. Support user update, deactivation/delete, role/status filtering, and guard-specific listings.
3. Implement scheduling, attendance, and workforce continuity automation.
    a. Create, update, delete, and list shifts with conflict-aware assignment.
    b. Enable guard check-in/check-out with attendance history tracking.
    c. Detect no-shows, request replacements, accept replacements, and manage guard availability.

4. Implement firearm inventory, issuance, and custody control.
    a. Provide full firearm CRUD with serial/model/caliber/status tracking.
    b. Implement firearm issuance, return, active allocations, and overdue allocation monitoring.
    c. Maintain firearm maintenance schedules, pending maintenance, and completion records.
5. Implement permit and training compliance management.
    a. Create and retrieve guard permits, including expiring and revoked permits.
    b. Support auto-expire permit processing for compliance enforcement.
    c. Manage training records, guard training history, and expiring training alerts.
6. Implement armored vehicle fleet and trip operations.
    a. Provide armored car CRUD, issuance/return, and active allocation monitoring.
    b. Manage driver assignment/unassignment and driver-vehicle linkage.
    c. Support trip creation, trip status updates, trip completion, and trip detail/history views.
    d. Manage vehicle maintenance scheduling, completion, and maintenance records.
7. Implement mission operations and merit-based performance evaluation.
    a. Enable mission assignment and mission retrieval for operational deployment.
    b. Calculate guard merit scores and provide ranked guard outputs.
    c. Support client evaluation submission, guard evaluation retrieval, and overtime candidate identification.
8. Implement support, communication, and notification services.
    a. Enable support ticket creation and guard ticket retrieval workflows.
    b. Provide user notifications, unread counts, mark-read/mark-all-read, and delete actions.

c. Support alerting for operational events and decision-critical updates.

9. Implement incident management and real-time tracking intelligence.
    a. Provide incident creation, active incident retrieval, and incident status updates.
    b. Implement guard heartbeat, tracking point capture, and map data endpoints.
    c. Provide client site CRUD and shift proximity alert checks.
    d. Enable real-time websocket tracking stream for live operational visibility.
10. Implement analytics, predictive intelligence, and decision support.
    a. Provide analytics overview, trends, and guard reliability endpoints.
    b. Implement predictive alerts for permit expiry, maintenance risk, no-show patterns, and guard capacity risk.
    c. Provide AI (deterministic) services for guard absence risk, replacement suggestions, vehicle maintenance risk, incident classification, and incident summarization.
11. Implement real-time location monitoring and map-based operations.
    a. Provide OpenStreetMap-based visualization of guards, routes, and client sites.
    b. Implement live tracking updates through websocket streaming.
    c. Support map data retrieval and proximity-based operational alerts.
12. Implement governance, security, and auditability mechanisms.
    a. Maintain write-request audit logs for critical actions.
    b. Support session/token-based authenticated API access and presence/last-seen tracking.
    c. Provide health-check endpoints for service monitoring and operational readiness.
13. Implement and validate cross-platform runtime delivery.
    a. Deliver the web runtime for browser-based command and operational access.

b. Deliver the desktop runtime through Tauri packaging for command-center deployment.
c. Deliver the mobile runtime through Capacitor Android packaging for field operations.
d. Validate cross-platform API connectivity and authentication/session behavior across Web, Desktop, and Android targets.

Scope and Limitations

Scope

The SENTINEL system is a web-based platform intended to strengthen operational control and compliance for a private security agency. The system is scoped to security agency operations and the regulatory environment set by the Private Security Services Industry Act (Republic Act No. 11917, 2022).

Data

- Personnel profiles, role/approval data, licensing and permit records, training records, schedules, attendance logs, availability status, and performance/merit metrics.
- Firearm inventory, firearm allocations, firearm maintenance records, and firearm permit compliance data.
- Armored vehicle assets, car allocations, driver assignments, trip records, and vehicle maintenance history.
- Missions, incidents, support tickets, notifications, predictive alerts, tracking/map points, and audit/access logs.
- Authentication lockout records, refresh-session lifecycle records, and audit source-IP traces for security monitoring and accountability.

Process

- Account lifecycle workflows including registration, email verification, authentication, password reset, and profile management.
- Approval workflows for newly registered guards before operational access is granted.
- Role-based dashboard workflows for command center, approvals, analytics, calendar operations, and audit-log monitoring.
- Shift scheduling, attendance validation, no-show detection, replacement request/acceptance, and supervisor oversight.
- Firearm issuance/return workflows with permit validation, maintenance scheduling, and custody traceability.
- Vehicle allocation, driver assignment, trip lifecycle management, and preventive/corrective maintenance tracking.
- Incident reporting, support ticket handling, notification delivery, and real-time operations tracking.
- Session hardening workflows including refresh-token rotation, logout revocation, and lockout persistence.

People

- Superadmin, Administrator, Supervisor, and Guard roles with distinct permissions and dashboard views

Technology

- Web-based application accessible on desktop and mobile browsers.
- Cross-platform runtime deployment for Web, Desktop (Tauri), and Mobile Android (Capacitor).
- React + TypeScript frontend and Rust + Axum backend services.
- Centralized PostgreSQL database with relational integrity, auditability,

and role-based data control.

- Dockerized deployment and API-driven architecture with websocket streaming and polling fallback for tracking views.

Limitation of the Study

- The system does not integrate with external payroll, HR, or government licensing systems.
- Dedicated hardware-grade vehicle telematics integration is not yet implemented; current tracking relies on application-provided location updates and available device/network conditions.
- AI/predictive outputs are decision-support recommendations and do not autonomously execute final operational actions.
- Direct integration with third-party hardware ecosystems (CCTV, IoT sensors, access control turnstiles) is not included.
- Offline-first field operation is limited; core workflows depend on network/API availability.
- Native wrapper scope is currently limited to Windows desktop and Android; iOS and macOS deployment targets are not included in the present implementation.

Review of Related Literature/Studies/Systems

Related Literature

The Philippine private security sector is currently navigating its most significant legal transition in over fifty years. The enactment of the Private Security Services Industry Act (Republic Act No. 11917, 2022) effectively repealed the outdated RA 5487, shifting the industry toward a regime of "Professionalized Accountability." According to Jur.ph (2025), the law's Implementing Rules and Regulations (IRR) mandate that Private Security Agencies (PSAs) maintain highly accurate, digitized records of License to Exercise Security Profession (LESP) and firearm permits. The act introduces a "strict liability" framework where administrative negligence, such as deploying an unlicensed guard, can result in fines ranging from ₱50,000 to ₱100,000 per violation, or up to ₱5,000,000 for agency-wide license failures.

This regulatory pressure has necessitated a move toward Automated Regulatory Compliance Tracking (ARCT). Research by Khinvasara, T., Shankar, A., & Wong, C. (2024) suggests that for organizations managing complex, shifting rules, the use of automated monitoring is a "significant advancement" that eliminates the "discovery lag" inherent in manual audits. By leveraging system-driven triggers for license expirations, PSAs can transition from reactive compliance to a proactive stance, ensuring that no personnel or asset is deployed without valid legal authorization.

Operational continuity in the private security sector is heavily dependent on reliable guard presence. As identified in the operational risks of DASIA Tagum, "No-Show" incidents represent a critical vulnerability where manual processes relying on phone calls and delayed human intervention fail to address "discovery lag." Research in workforce capacity

optimization emphasizes that manual, reactive responses to workforce gaps significantly increase "vacancy time," directly correlating with higher security breach probabilities (Shiyanbola et al., 2023).

Modern systems mitigate this by automating the identification and deployment of replacements based on real-time availability, shifting the organizational stance from reactive crisis management to proactive risk mitigation (Al-Khafajiy et al., 2022). Furthermore, advanced algorithms now enable "predictive scheduling," which analyzes historical No-Show data to alert administrators of potential gaps before they occur, drastically reducing the operational vacancy time (TrackTik, 2025).

The transition from inconsistent, manual tracking to a structured, data-driven framework is essential for maintaining Distributive Justice, the perceived fairness of how shifts, rewards, and responsibilities are allocated within an organization. Aguinis (2022) posits that the absence of a structured performance management framework leads to "Social Loafing," a psychological phenomenon where employees decrease effort due to a perceived lack of individual accountability.

For security agencies like DASIA Tagum, implementing centralized performance metrics such as punctuality and client feedback logs ensures that shift allocation is based on objective data rather than subjective selection. This transparency not only increases overall guard reliability but also fosters a culture of accountability, as guards understand that their performance is directly linked to future opportunities (Tan et al., 2024).

The systematic allocation of firearms is a matter of both Chain of Custody (CoC) and legal necessity under the Private Security Services Industry Act (Republic Act No. 11917, 2022). The law mandates strict accountability for firearms, requiring that only authorized, trained, and compliant guards carry weapons. Manual processes often suffer from "data drift," where the status of a weapon and the permit validity of the assigned guard are not reconciled in real-time, leading to potential administrative negligence. Centralized digital tracking is identified as the primary defense against the misallocation of high-risk equipment and is a prerequisite for generating the legally defensible audit logs required by the National Cybersecurity Plan 2022-2028 (DICT, 2024; PNP-SOSIA, 2022).

The selection of the backend technical stack is a core security decision for mission-critical systems. Scofield, M. B. (2025) argues that Rust's "ownership and borrowing" model provides a fundamental architectural advantage: Memory Safety without a Garbage Collector. This ensures that SENTINEL is natively resistant to buffer overflows and data races, vulnerabilities that often plague systems handling high-concurrency data like real-time firearm tracking.

Complementing the language is the Axum framework, which is built on the Tokio asynchronous runtime. As highlighted by NashTech (2025), Axum is designed for high-performance web services where scalability and raw speed are non-negotiable. For SENTINEL, this means the Attendance Tracking and No-Show Detection modules can handle hundreds of simultaneous "check-in" requests during shift rotations without performance degradation. This "asynchronous-first" design allows the system to remain responsive even when processing complex relational logic across personnel, vehicle, and firearm databases.

In an Integrated Operations Platform, the database must serve as the immutable "Single Source of Truth." PostgreSQL (2025) is recognized as the premier open-source RDBMS for enterprise planning due to its strict adherence to ACID (Atomicity, Consistency, Isolation, Durability) properties. Nguyen, R. (2025) emphasizes that PostgreSQL's implementation of Foreign Key Constraints and Unique Exclusion Constraints is the primary defense against "data drift."

In the SENTINEL ecosystem, these constraints ensure that a firearm or armored vehicle cannot be logically "double-booked" or assigned to a guard who does not exist in the personnel table. This high level of relational integrity is a prerequisite for generating the legally defensible Audit Logs required by the National Cybersecurity Plan 2023-2028 (DICT, 2024), ensuring that every asset movement is tied to a verifiable digital identity.

Related Studies and/or Systems

Empirical research confirms that automated oversight is a primary driver of operational integrity. A study by Atlam, H. F., & Yang, Y. (2025) found that organizations utilizing unified Access Control and Resource Planning (ERP) systems saw a 27% drop in unauthorized access violations and reclaimed 36% of staff time previously lost to manual verification. These findings suggest that hardwiring compliance into the system architecture, rather than treating it as an afterthought, tightens process integrity and reduces the risk of internal data tampering.

Furthermore, research by Shiyanbola, J. O., et al. (2023) on "Workforce Capacity Optimization" highlights a flaw in traditional security: "discovery lag." This occurs when supervisors only identify a personnel gap after a shift has failed. Their study demonstrates that real-time, asynchronous detection models allow for a proactive 11.8% improvement in site coverage reliability. Locally, ResearchGate (2025) cites a study by Respicio (2023) which found that Philippine security guards operating under digital monitoring exhibited significantly higher levels of alertness and punctuality because the system provided a transparent, inescapable layer of accountability that manual logs could not match. The functional blueprint of SENTINEL is informed by the performance of several industry-leading platforms such as TrackTik and Guardhouse.

TrackTik

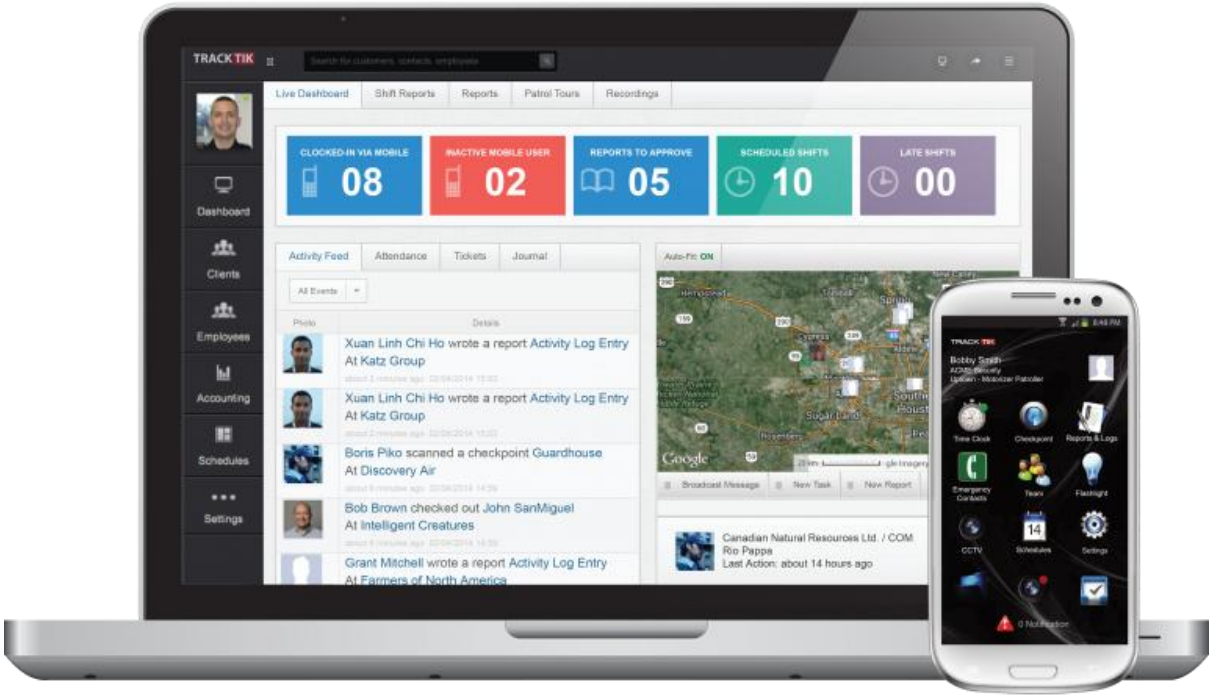

Figure 1. TrackTik

TrackTik (2025) has redefined security workforce management by moving beyond static scheduling toward an integrated, AI-driven operational model. According to the Trackforce 2025 Physical Security Operations Benchmark Report, the industry is currently facing an "AI adoption paradox" where the value of predictive scheduling is recognized, but cost remains a barrier for mid-sized firms. TrackTik's success is anchored in its ability to consolidate alarm monitoring, guard dispatch, and business administration into a single 24/7 resilient architecture, which users report has increased operational efficiency by approximately 20%. This efficiency is primarily attributed to the reduction of manual administrative burdens through automated compliance tracking and real-time intelligence feeds (TrackTik, 2025).

Guardhouse

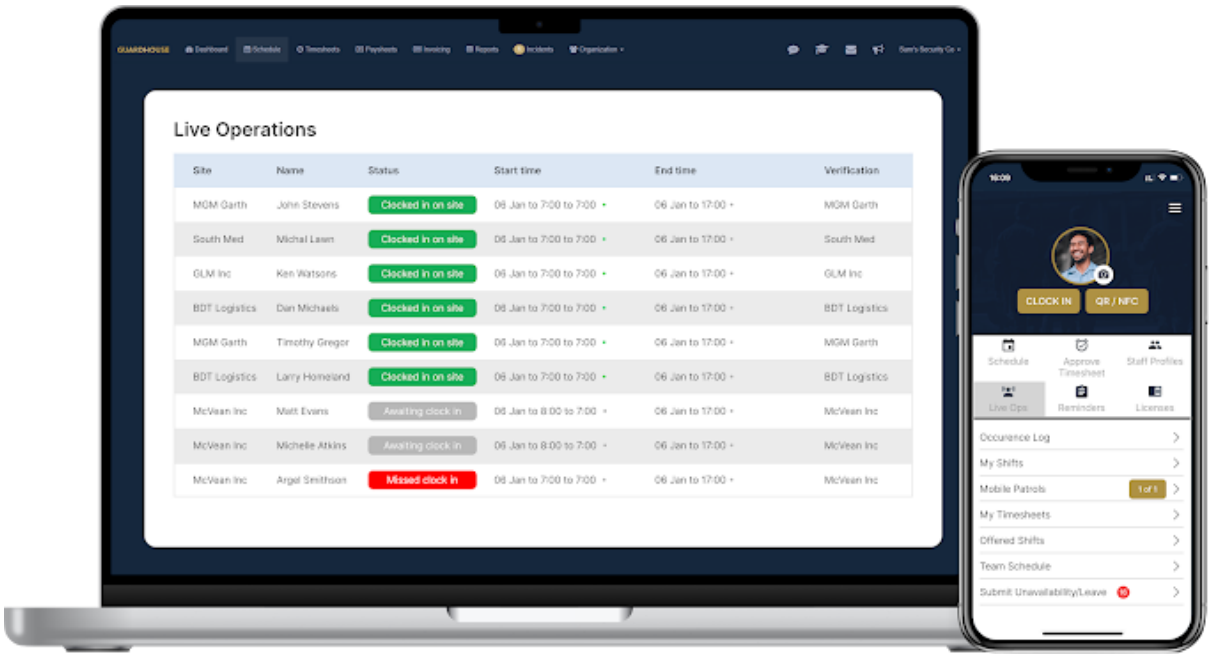

Figure 2. Guardhouse

Guardhouse (2026) addresses the specific administrative "overload" that has become a measurable barrier to performance in the mid-2020s. Research from NODE Magazine (2026) indicates that security employees often lose an average of 15 hours per week to routine administrative tasks, such as re-entering data across fragmented systems. Guardhouse mitigates this by unifying scheduling, GPS tracking, and invoicing into a streamlined workflow, reportedly reducing office-based administrative time by 40–60%. Its "Confidence in Compliance" module provides a specialized framework for daily license verification, a logic that SENTINEL adapts to specifically meet the stringent requirements of RA 11917 (GetApp, 2026).

MySecuritas

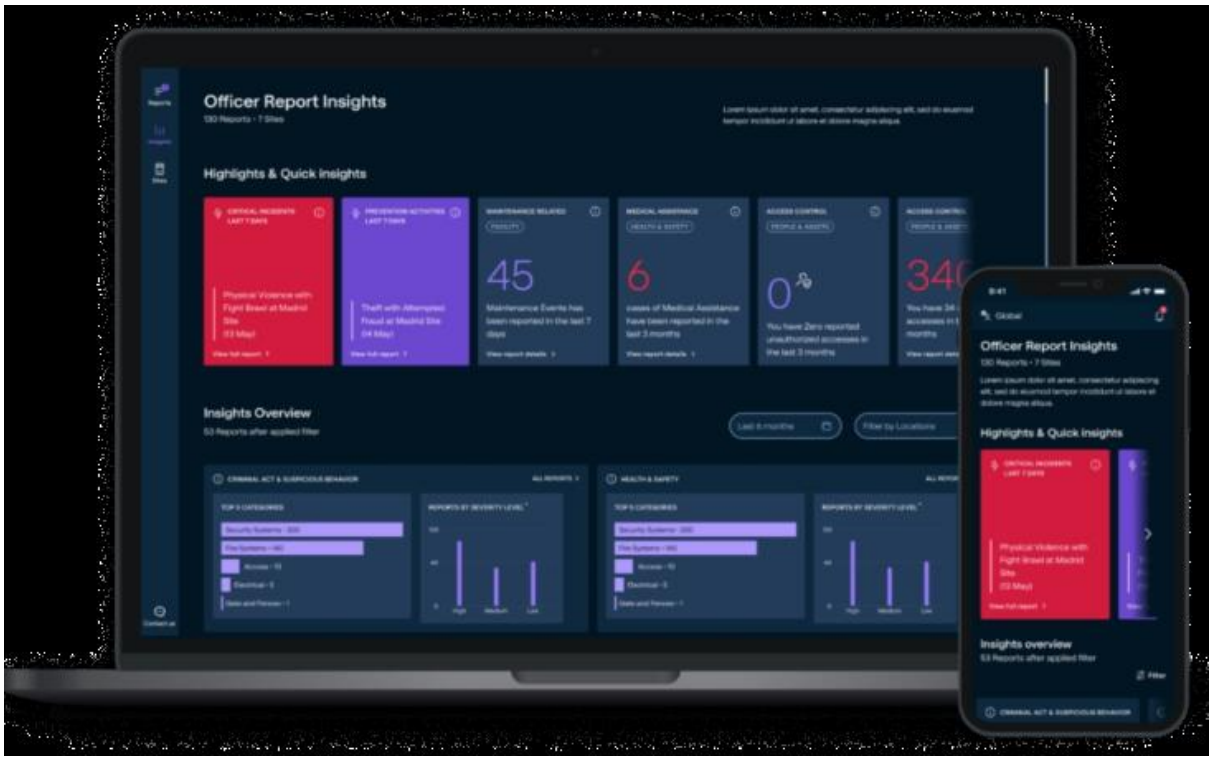

Figure 3. MySecuritas

Securitas "MySecuritas" (2026) represents the industry's shift toward "Situational Understanding" and proactive risk management. The Securitas 2026 Global Technology Outlook Report identifies that the rapid advancement of generative AI for contextual understanding is now a top priority for 30% of security decision-makers (Securitas Technology, 2025). MySecuritas operationalizes this by providing a unified dashboard that transforms disparate incident reports into actionable statistics and trend highlights. This allows supervisors to identify risk patterns before they escalate, a core design principle mirrored in SENTINEL's Performance Analytics Dashboard (Securitas, 2026).

Silvertrac Software

Figure 4. Silvertrac Software

Silvertrac Software (2026) is widely utilized for "Proof of Performance" and on-site accountability, particularly in parking and property management environments. While it excels in field reporting and checkpoint verification via mobile applications, its architectural focus remains on incident management and officer accountability (Silvertrac Software, 2026). Technical evaluations of generic guard management systems suggest a persistent "visibility gap" in specialized asset tracking; most platforms lack the deep relational logic required for a Digital Chain of Custody for high-risk assets like firearms and armored car fleets (Hardcat, 2025). SENTINEL fills this niche by integrating these high-stakes logistics, traditionally managed in siloed armory or fleet software, into the primary workforce management ecosystem.

Across the reviewed literature, several consistent themes emerge. Researchers emphasize the importance of integrating operational data to improve organizational visibility and decision-making. Studies from both Philippine and international contexts highlight the limitations of fragmented monitoring tools and manual processes.

Technological trends such as automated scheduling algorithms, attendance tracking, and operational dashboards are increasingly adopted across industries. However, existing systems often address only a single domain of operations, such as workforce scheduling or surveillance monitoring. Global systems like TrackTik and Guardhouse demonstrate that the industry standard is defined by Real-Time Accountability and Centralized Data Ownership. However, these systems often fail to accommodate the granular administrative nuances of Philippine law.

The reviewed literature therefore reveals a clear research gap in integrated systems that combine workforce management, compliance monitoring, and operational oversight within a unified platform tailored to security agencies. The SENTINEL system addresses this void by combining high-concurrency Rust/Axum architecture with modules specifically for Philippine-specific licensing (RA 11917), Firearm Custody, and Armored Car Fleet Management, allowing local agencies to achieve multinational levels of visibility and compliance. It provides a centralized solution that integrates scheduling automation, regulatory compliance verification, and real-time operational dashboards within a single enterprise system.

METHODOLOGY

The development methodology used for SENTINEL is iterative and Agile-inspired, structured around feature-based increments rather than a single linear build. Each increment combined planning, implementation, integration, and review so that operational modules could be delivered and validated in usable slices. This approach was selected because SENTINEL includes tightly coupled operational domains (personnel, assets, tracking, incidents, and analytics) that require frequent cross-module verification.

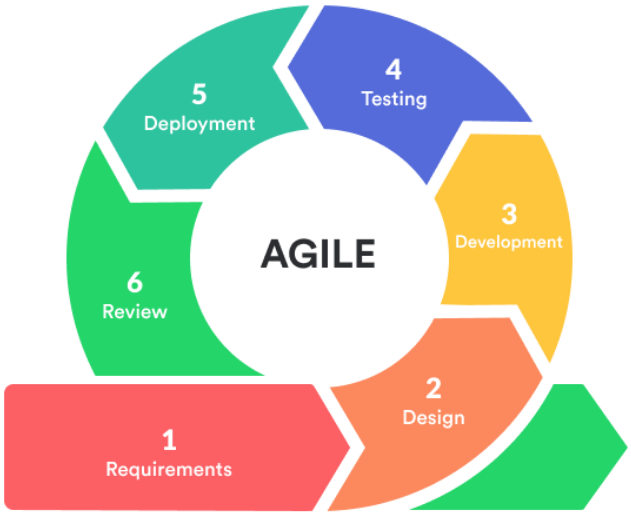

Figure 5. Agile Model

Planning activities began with requirements consolidation, role-mapping, and module prioritization. The team translated these into implementation batches covering authentication and approvals, scheduling and attendance, firearms and fleet operations, incident/support workflows, analytics dashboards, tracking, and AI-assisted operational intelligence. Platform targets (web, desktop, and mobile) were included in planning to prevent late-stage portability redesign.

Development was executed through frontend-backend parallel work with recurring integration checkpoints. Frontend components and hooks were built alongside backend

handlers, middleware, and service logic, then connected through authenticated API contracts. This feature-based integration pattern reduced interface drift and enabled earlier discovery of role-permission, payload, and workflow defects.

Testing and refinement were continuous throughout implementation. The team performed module-level checks, cross-role scenario tests, API contract verification, and runtime build validation for web, desktop, and Android outputs. Security and reliability refinements (for example lockout persistence, refresh-session revocation, authorization enforcement, and rate-limiting controls) were integrated as iterative hardening tasks.

Deployment considerations were addressed through Dockerized service orchestration, PostgreSQL-backed persistence, and release-oriented build scripts for cross-platform distribution. This ensured that the methodology did not end at coding, but covered operational readiness and maintainability for real-world SOC usage.

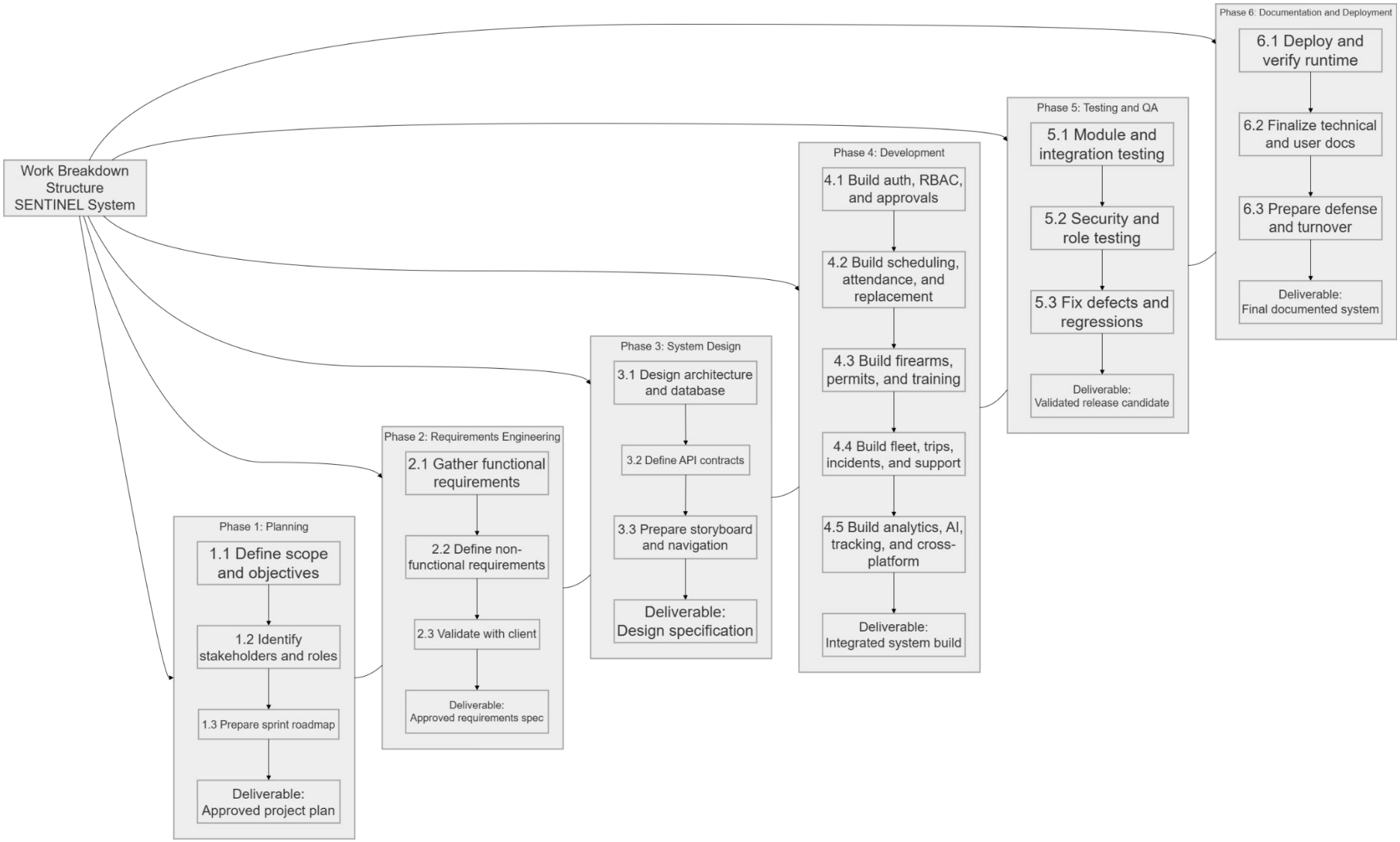

Figure 6. Work Breakdown Structure

Technical Background

Technologies to be used in the System

SENTINEL is implemented as a web-first integrated security operations platform with desktop and mobile runtime wrappers. Technology selection was based on concrete implementation requirements: role-based dashboards, secure API contracts, high-concurrency backend operations, traceable relational data, real-time tracking, and cross-platform delivery without codebase duplication. The proponents used the following technologies in the current implementation:

1. **React + TypeScript:** Will be used to build reusable, role-aware UI components and dashboard modules with strict typing.
2. **Vite:** Will be used for fast local startup and production build pipelines.
3. **Tailwind CSS:** Will be used for utility-driven, responsive interface implementation across command center, operations, and resource modules.
4. **Leaflet + OpenStreetMap:** Will be used for operational map rendering, live marker visualization, and client-site management.
5. **Rust + Axum:** Will be used for backend API handlers, middleware, and service orchestration.
6. **PostgreSQL:** Will be sed as the central transactional database for users, approvals, schedules, attendance, firearms, vehicles, incidents, notifications, tracking points, AI outputs, and audit logs.
7. **Docker + Docker Compose:** Will be ed to standardize local backend/database orchestration and deployment parity.
8. **JWT Authentication:** Will be used for role-scoped API access, session continuity, refresh rotation, and logout revocation.

9. **WebSocket + polling fallback:** Will be used for real-time tracking snapshots with continuity under unstable transport conditions.
10. **Capacitor:** Will be used to package the web frontend for Android field deployment.
11. **Tauri:** Will be used to package the same frontend for Windows desktop command-center deployment.
12. **Resend:** Will be used for transactional email delivery such as verification codes, password reset codes, and operational notifications.
13. **Railway:** Will be used as a cloud deployment platform for backend runtime and managed service hosting.
14. **Namecheap:** Will be used for domain registration and domain-level DNS management.
15. **Git and GitHub:** Will be used for version control, collaboration, and change tracking.

Gantt Chart of Activities

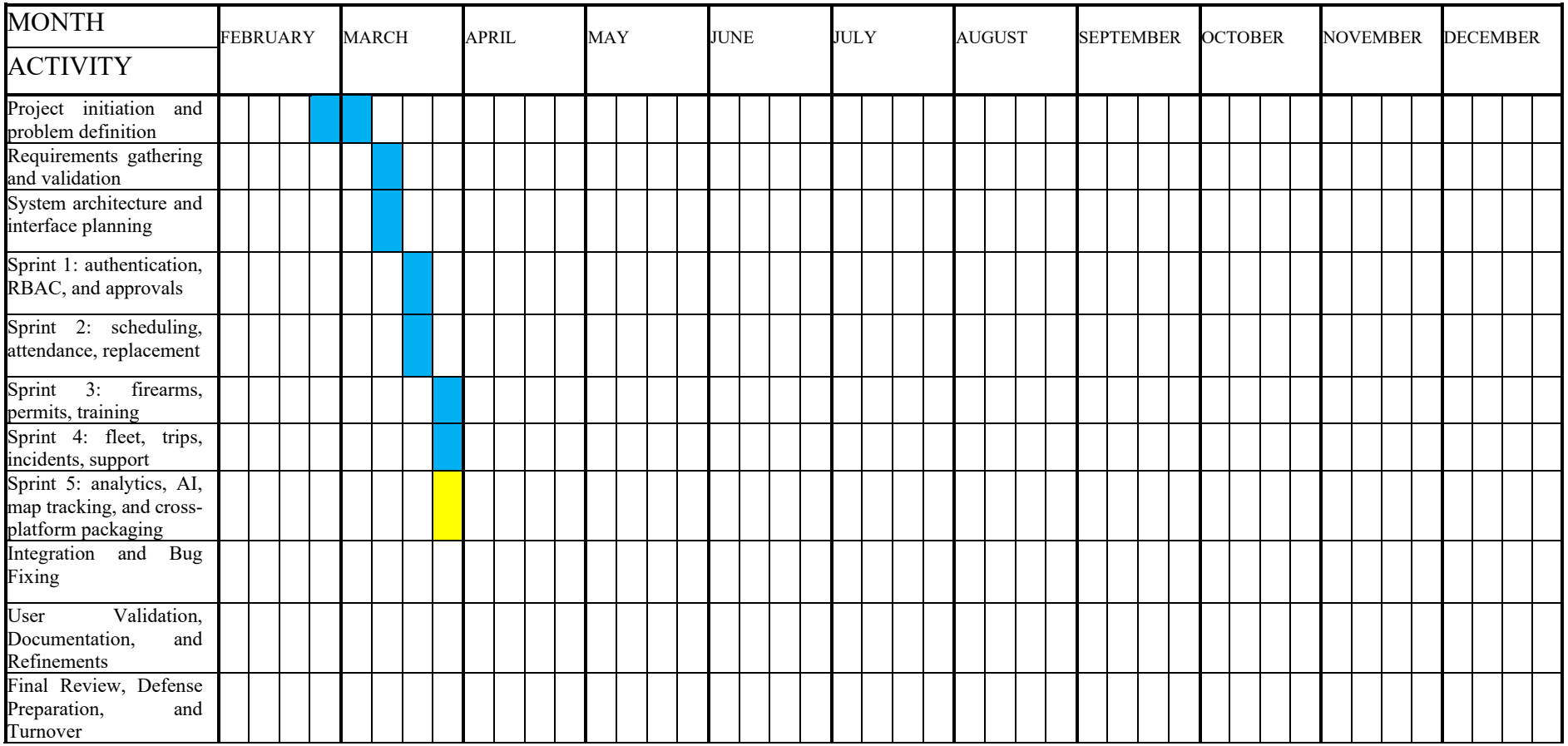

Figure 7. Gantt Chart of Activities

Legends: Completed Ongoing Not Started

Calendar of Activities

Figure 7 represents the calendar of activities that summarizes the tasks and their duration. Activities are listed on the left, and the schedule is at the top right. Bars indicate task progress, with each bar representing a week. The calendar of activities summarizes the same phased sequence defined in the WBS and Gantt schedule, including objectives, involved personnel, and required resources.

Resources

Hardware Requirements (Recommended)

Hardware specifications play a crucial role in the development of a web-based platform. The specific requirements will vary based on the proponents' needs and the expected workload. The following are the minimum requirements for computer hardware and laptops.

Table 1: Minimum System Requirements of the Laptop or Computer

Table 1 presents the minimum hardware requirements for the SENTINEL system to function effectively. These specifications are a baseline to ensure smooth system operation during development, deployment, and daily use.

Hardware	Specifications
Operating Systems	Windows 10 or Windows 11
Processor	Ryzen 3 or equivalent
Memory / RAM	8 GB minimum (16 GB recommended)
Disk Space / Disc Storage	250GB SSD
Graphics	Integrated Graphics
Screen Resolution	1366x768 or higher
Connection	Requires broadband internet connection

Deployment Environment

Application Server	4 vCPU minimum, 8-16 GB RAM, SSD-backed storage, Docker-capable host
Database Server	PostgreSQL-capable host with regular backup storage
Command Center Endpoints	Windows desktop systems capable of running Tauri desktop build
Field Endpoints	Android smartphones/tablets with GPS and reliable mobile data/Wi-Fi connectivity for guard tracking and attendance workflows

Software

The following software requirements are necessary for the development of the project and are related to system development, testing, deployment, documentation, and collaboration tools used by the proponents:

Development Requirements

1. Visual Studio Code for frontend, backend, and documentation workflow management.
2. Node.js and npm for React/Vite dependency management, development server execution, and production frontend builds.
3. Rust toolchain (rustup, cargo) for building and validating the Axum backend services.
4. Docker Desktop and Docker Compose for local orchestration of backend and PostgreSQL services.
5. PostgreSQL tooling (local or containerized) for schema validation, data checks, and operational query verification.
6. Git and GitHub for version control, branch-based collaboration, and release workflow integration.

Runtime Requirements

7. Modern web browser (Chromium-based or equivalent) for the web deployment target.
8. Windows desktop operating system for the Tauri-based desktop deployment target.
9. Android operating system for the Capacitor-based mobile deployment target.
10. Network connectivity between client runtimes and backend API services, including websocket access for live tracking views.

Requirements Analysis

The requirements baseline for SENTINEL was defined from actual private-security operational pain points: fragmented coordination, delayed visibility, compliance exposure, and weak traceability of sensitive asset workflows. Based on current implementation, the requirements analysis prioritizes role-governed access, real-time operational awareness, and auditable process execution across personnel, equipment, vehicle, incident, and analytics domains.

Technology and software selection was treated as a requirements decision, not only an implementation preference. For the frontend layer, React + TypeScript + Vite was selected to satisfy modular dashboard composition, strict role-based rendering, and rapid iteration. Angular was considered because it provides a full framework with built-in dependency injection and strong conventions; however, its heavier project structure was less aligned with the team's component-by-component delivery pace. Vue was considered for its simpler learning curve and progressive adoption model, but the existing codebase and shared component strategy were already established around React patterns. Plain JavaScript React was also considered, but TypeScript was chosen to reduce contract errors in role logic, API payload mapping, and cross-module state handling.

For backend processing, Rust + Axum was selected to meet concurrency, security, and reliability requirements for authentication, approvals, tracking, and analytics endpoints. Node.js + Express was considered because of rapid API prototyping and broad ecosystem support, but the project prioritized compile-time safety and stricter runtime guarantees for security-sensitive operations. Java + Spring Boot was considered for enterprise-grade structure and mature tooling, yet it introduces a larger operational footprint and greater configuration overhead for this capstone scope. Go + Gin/Fiber was also considered for performance and simplicity, but Rust + Axum better matched the team's objective of memory-safe, middleware-centric API enforcement with fine-grained control.

For persistence, PostgreSQL was selected because SENTINEL requires relational integrity across users, approvals, shifts, attendance, firearms, vehicles, incidents, and audit records. MySQL/MariaDB were considered because they are widely used relational systems, but PostgreSQL was preferred for stronger advanced SQL features, robust JSONB support, and flexible indexing for mixed transactional and analytical workloads. MongoDB was considered because of schema flexibility and rapid document modeling; however, SENTINEL's compliance and cross-entity constraints require strong relational guarantees that are more naturally enforced in PostgreSQL.

For mapping and geospatial visualization, OpenStreetMap + Leaflet was selected to satisfy real-time operational map requirements, client-site management, and location marker rendering while retaining implementation control. Google Maps Platform was considered because it provides high-quality basemaps, geocoding, and route services, but usage-based billing and API quota sensitivity are less favorable for continuous monitoring views. Mapbox was considered because it offers strong vector map tooling and customizable styles, yet recurring usage costs and token-governed access create additional operational dependency for long-running dashboards. ArcGIS was considered for enterprise GIS depth, but its platform

complexity exceeds current implementation needs. OpenStreetMap + Leaflet was therefore chosen for cost efficiency, integration flexibility, and direct alignment with the implemented web map architecture.

For real-time operational visibility, websocket streaming with polling fallback was selected as a hybrid requirement. Server-Sent Events (SSE) was considered because it simplifies one-way streaming, but websocket transport better supports bidirectional session handling and live snapshot exchange patterns used in tracking views. Polling-only models were considered for implementation simplicity, but they introduce higher latency and unnecessary repeated requests for active command-center monitoring. The implemented websocket-plus-polling model balances responsiveness with continuity when persistent connections are interrupted.

For deployment targets, the system uses a web-first core with Tauri (desktop) and Capacitor (Android) wrappers to meet cross-platform access requirements without maintaining separate business logic per platform. Electron was considered for desktop packaging because of mature tooling, but Tauri was preferred for a lighter runtime footprint and tighter security posture. React Native and fully native Android/desktop tracks were considered for platform-specific delivery, but they would require duplicate UI and integration maintenance. The chosen wrapper strategy preserves feature parity across Web, Desktop, and Android while reducing implementation divergence.

For operational tooling, Docker/Compose, managed runtime deployment, and version-controlled release workflows were selected to satisfy reproducibility and maintainability requirements. Manual host-based installation was considered but increases environment drift and onboarding inconsistency. Alternative orchestration-heavy stacks (for example, immediate Kubernetes adoption) were considered excessive for current scale and capstone constraints.

The selected tooling stack provides repeatable setup, traceable change history, and controlled release validation that directly supports the implemented system lifecycle.

Requirements Documentation

Functional Requirements

- The system shall support account registration, verification, approval, authentication, and password-reset workflows for operational users.
- The system shall enforce role-based access for superadmin, admin, supervisor, and guard, including role-appropriate dashboard and API access.
- The system shall support guard management workflows including profile governance, shift assignment, attendance capture, no-show detection, and replacement processing.
- The system shall support firearm lifecycle management including records, issuance/return, permit-state awareness, and maintenance tracking.
- The system shall support armored vehicle lifecycle management including records, driver assignment, trip operations, and maintenance tracking.
- The system shall provide real-time tracking capabilities through map data endpoints, heartbeat ingestion, client-site management, and live websocket updates with polling fallback.
- The system shall provide analytics and command-level dashboards for operational summaries, trends, approvals, and decision support.
- The system shall provide AI-assisted decision-support outputs for absence risk, replacement recommendation, incident classification, incident summarization, and predictive alerts.

- The system shall provide ticketing and notification workflows to support operational communication and exception handling.
- The system shall support cross-platform runtime delivery for web, Windows desktop (Tauri), and Android mobile (Capacitor).
- The system shall provide audit-log visibility for authorized elevated roles, with filterable and paginated audit records.

Non-functional Requirements

- Security: The system shall enforce JWT-based authentication, RBAC authorization, approval-gated guard access, audit logging, lockout controls, and rate limiting for abuse resistance.
- Performance: The system shall maintain responsive dashboard behavior and near-real-time update delivery via websocket and timed polling refresh cycles.
- Usability: The system shall provide role-centered dashboards and modular UI navigation that reduce operator context switching during active operations.
- Reliability: The system shall support robust session continuity through refresh-token persistence, rotation, and explicit logout revocation.
- Scalability: The system architecture shall remain API-driven and modular so additional modules, integrations, and platform targets can be added without redesigning the full stack.

Design of Software, System, Product, and/or Processes

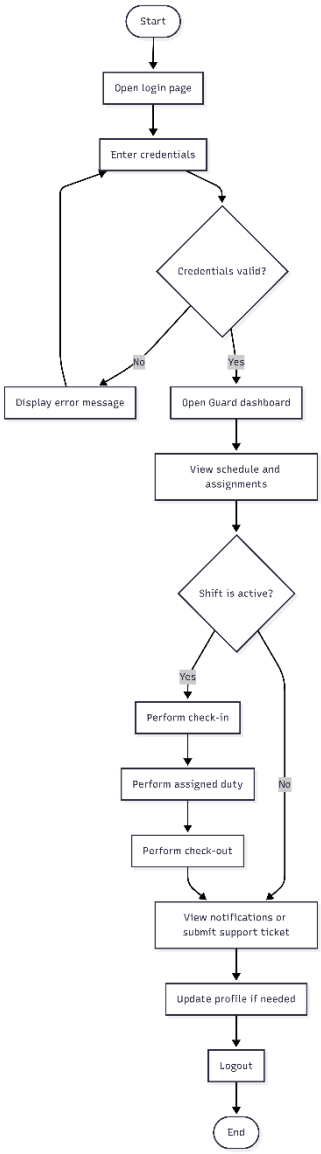

Figure 8. Guard Activity Diagram

Figure 8 illustrates the workflow of a Guard within the proposed system, beginning with account login and credential validation. After successful authentication, the Guard will access the role-based dashboard to view assigned schedules and duty details. When a shift is active, the Guard will perform check-in, execute assigned responsibilities, and complete check-out. If no active shift is detected, the Guard can still review notifications, submit support concerns, and update profile details as needed. The process will end with secure logout, ensuring that daily field operations and account access are properly closed.

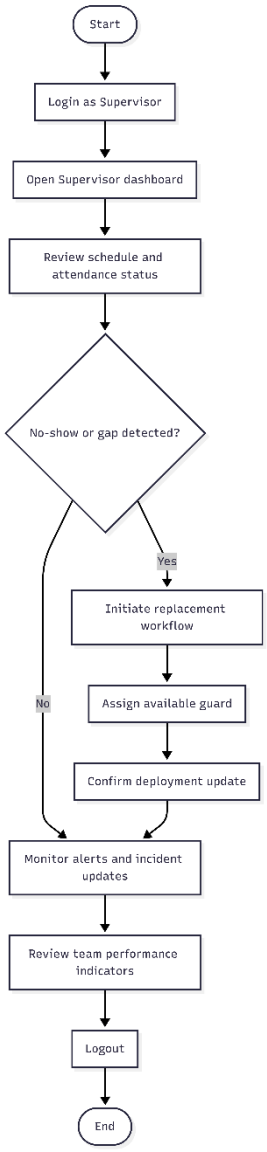

Figure 9. Supervisor Activity Diagram

Figure 9 presents the process flow for the Supervisor role, starting from login and access to the supervision dashboard. The Supervisor will review schedules and attendance to detect no-shows or staffing gaps. When operational gaps are identified, the system will guide the Supervisor through replacement assignment and deployment confirmation. If no gap is found, the Supervisor can continue monitoring alerts, incidents, and team performance indicators. The workflow emphasizes timely intervention, operational continuity, and controlled session closure through logout.

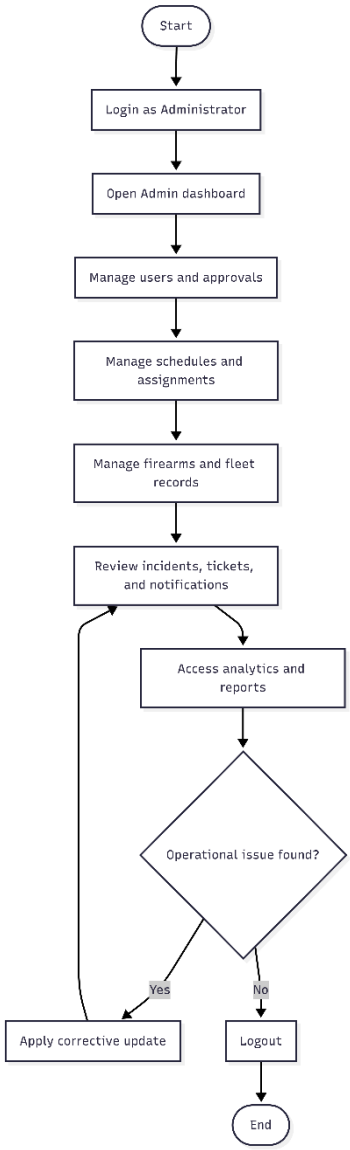

Figure 10. Administrator Activity Diagram

Figure 10 outlines the activities of the Administrator, who manages core operational modules after authentication. From the Admin dashboard, the Administrator can process user approvals, manage schedules, and maintain firearm and fleet records. The role also includes reviewing incidents, support tickets, notifications, and analytics outputs for decision support. When an operational issue is detected, corrective actions can be applied and revalidated through the same workflow loop. The sequence ends with logout to preserve account security and traceability.

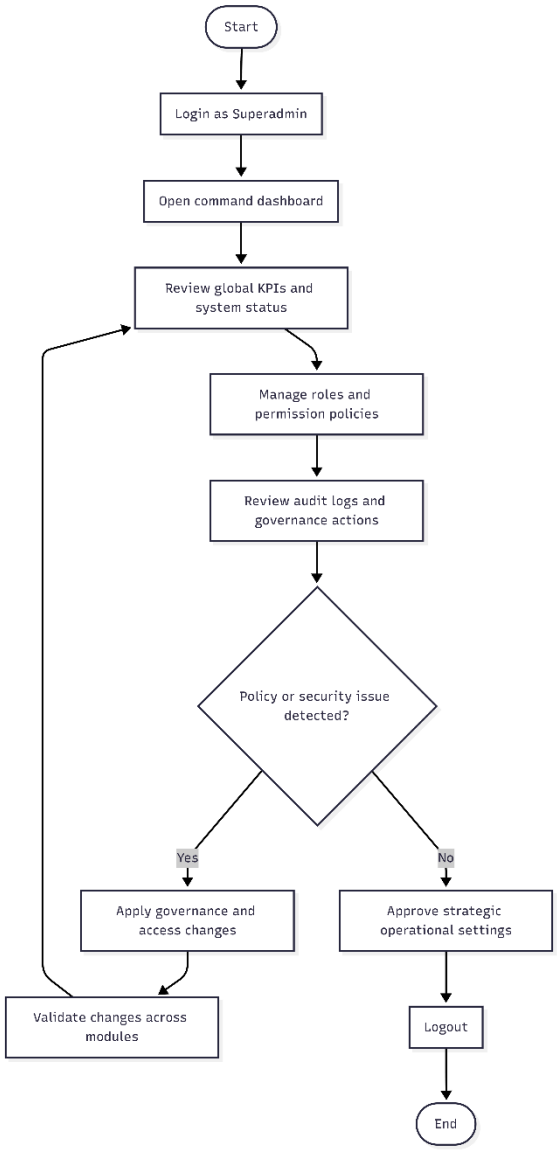

Figure 11. Superadmin Activity Diagram

Figure 11 describes the Superadmin workflow, which focuses on system-wide governance and strategic control. After login, the Superadmin will review global performance indicators, enforce role and permission policies, and inspect audit logs for security and compliance visibility. If policy or security concerns are detected, governance settings and access controls can be updated and validated across modules before resuming monitoring. If no issue is found, strategic approvals can proceed before session termination. This design supports high-level oversight, accountability, and secure administration across the entire platform.

Development

The development component of SENTINEL will be organized into functional modules so that each operational requirement can be implemented, tested, and validated systematically.

- User Login Module

  This module will enable secure access to the system through role-based authentication. The login process will support Superadmin, Administrator, Supervisor, and Guard accounts, each with role-scoped dashboard access and permissions. It will also support account verification and password reset workflows.

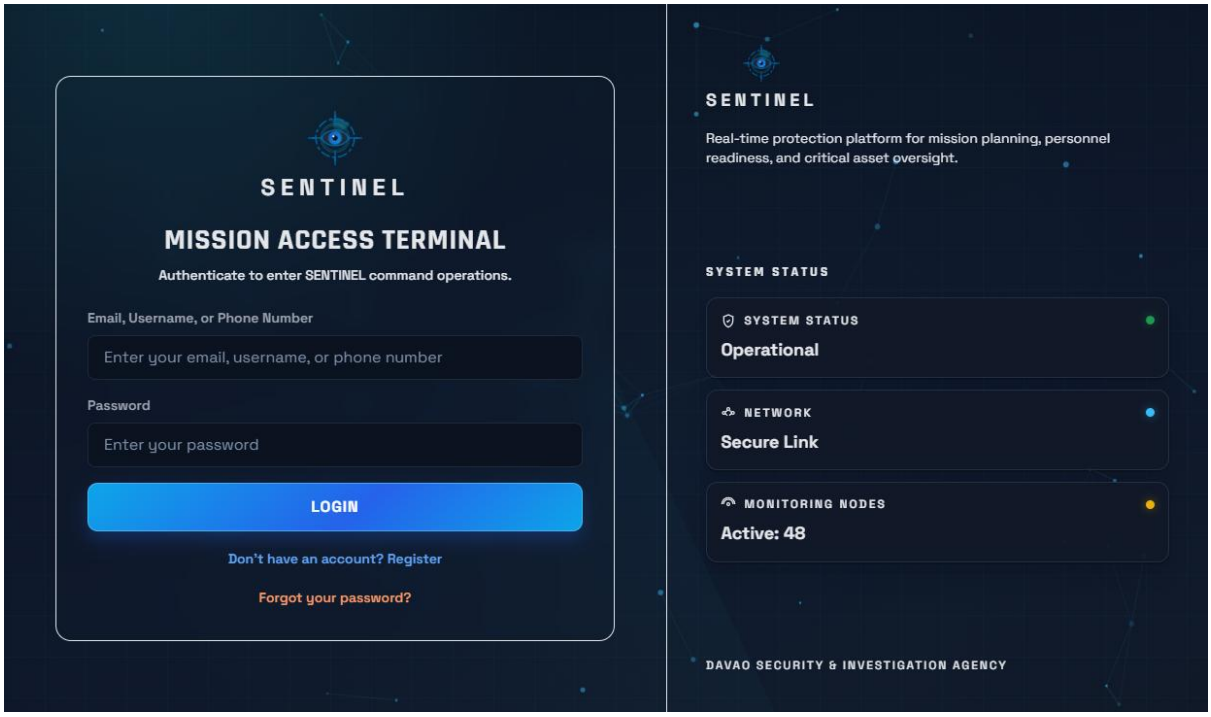

Figure 12. User Login Module

- Superadmin Module

  This module will serve as the system governance layer for platform-wide control. The Superadmin will manage global role policies, permission structures, audit visibility, and strategic operational settings to maintain security and compliance across all modules.

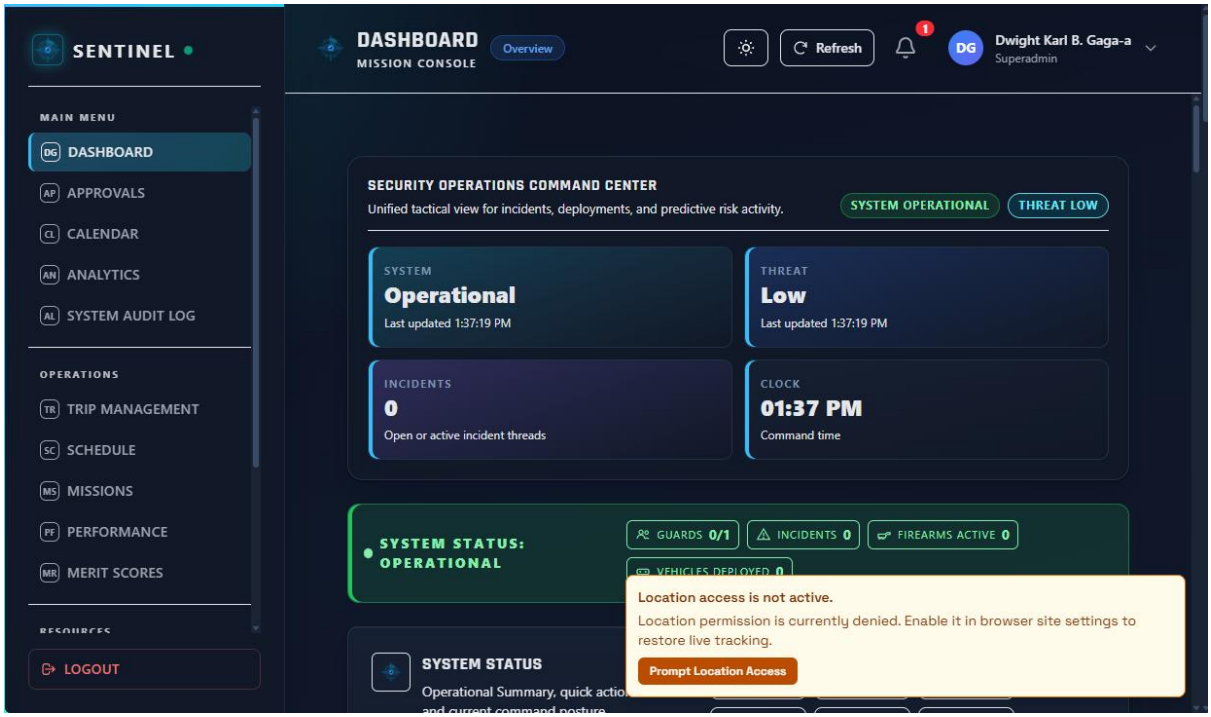

Figure 13. Superadmin Module

- Administrator Module

This module will serve as the operations management gateway for administrative users. Administrators will manage users, approvals, schedules, asset records, compliance entries, notifications, and system-level operational updates, with access to reporting and analytics views.

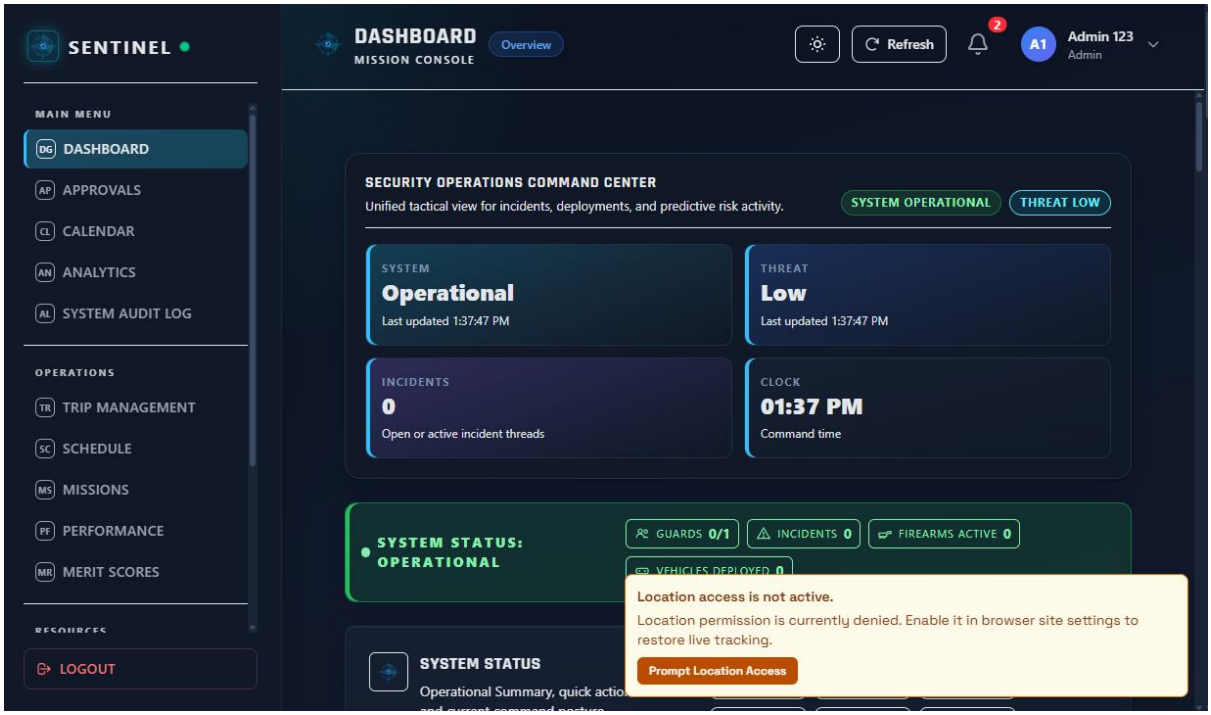

Figure 14. Administrator Module

- Supervisor Module

This module will support day-to-day field supervision and deployment continuity. Supervisors will monitor attendance and schedules, detect no-shows, initiate replacements, track incidents and alerts, and review duty coverage performance.

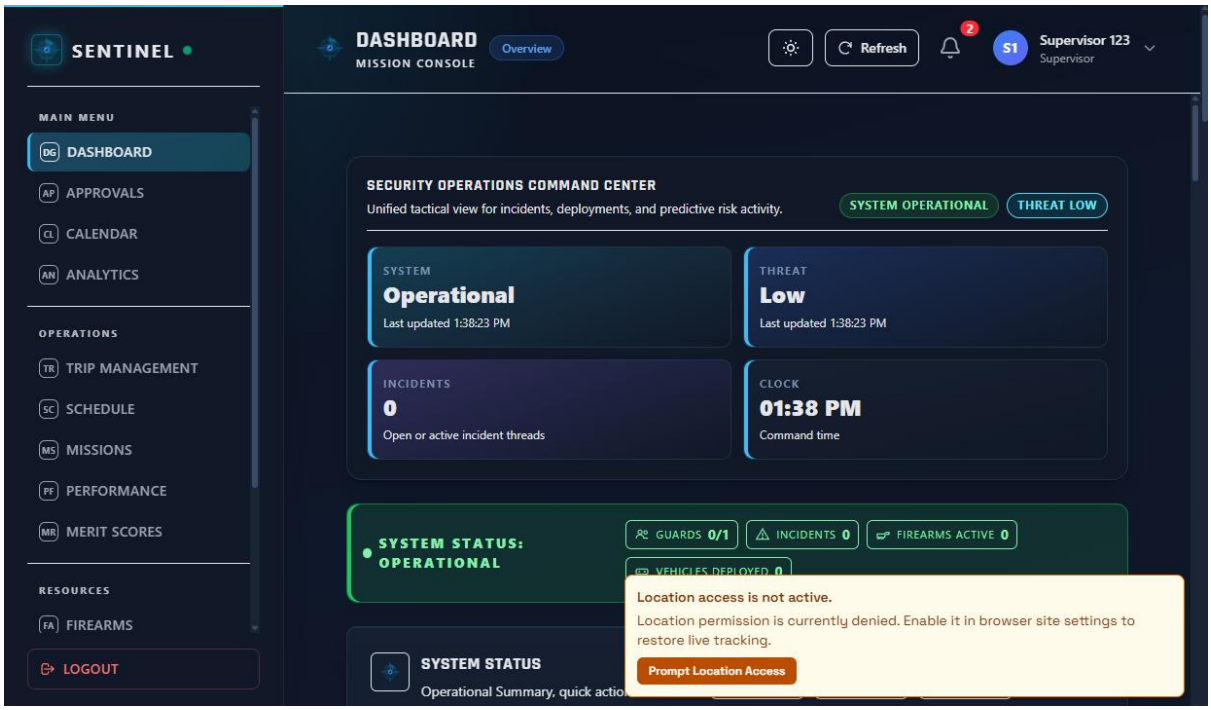

Figure 15. Supervisor Module

- Guard Module

This module will serve as the field-user workspace for execution of assigned duties. Guards will view schedules and assignments, perform check-in and check-out, review notifications, submit support tickets, and manage allowed profile updates.

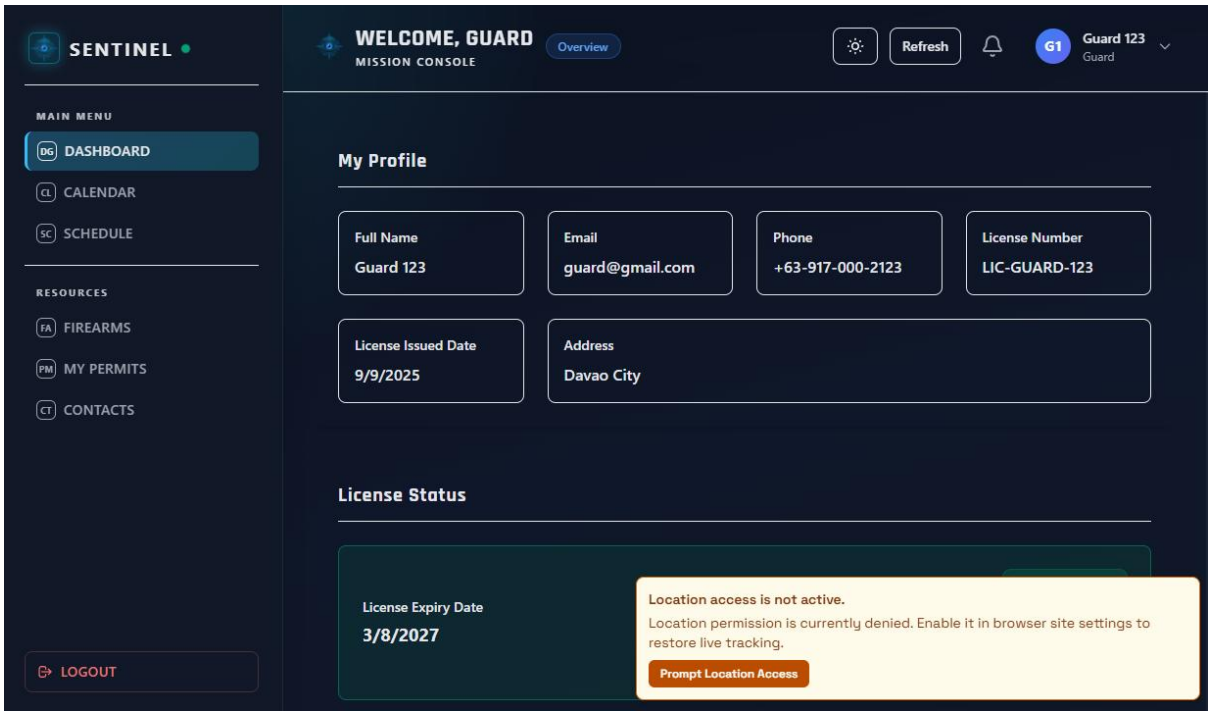

Figure 16. Guard Module

- Scheduling and Attendance Module

This module will handle shift planning, assignment mapping, attendance capture, and no-show detection. It will coordinate replacement workflows and provide supervisors and administrators with real-time staffing visibility.

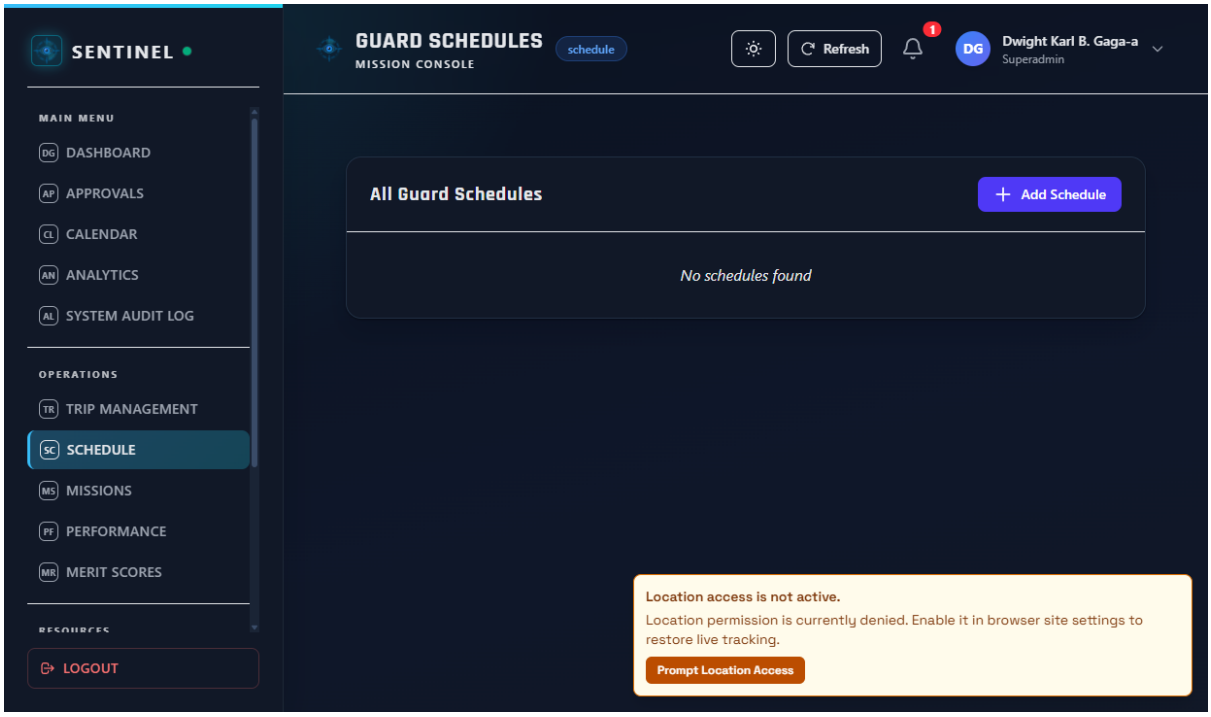

Figure 17. Scheduling and Attendance Module

- Asset and Compliance Module

This module will centralize firearm records, permit validity, maintenance entries, armored vehicle tracking, and related compliance controls aligned with RA 11917 operational requirements.

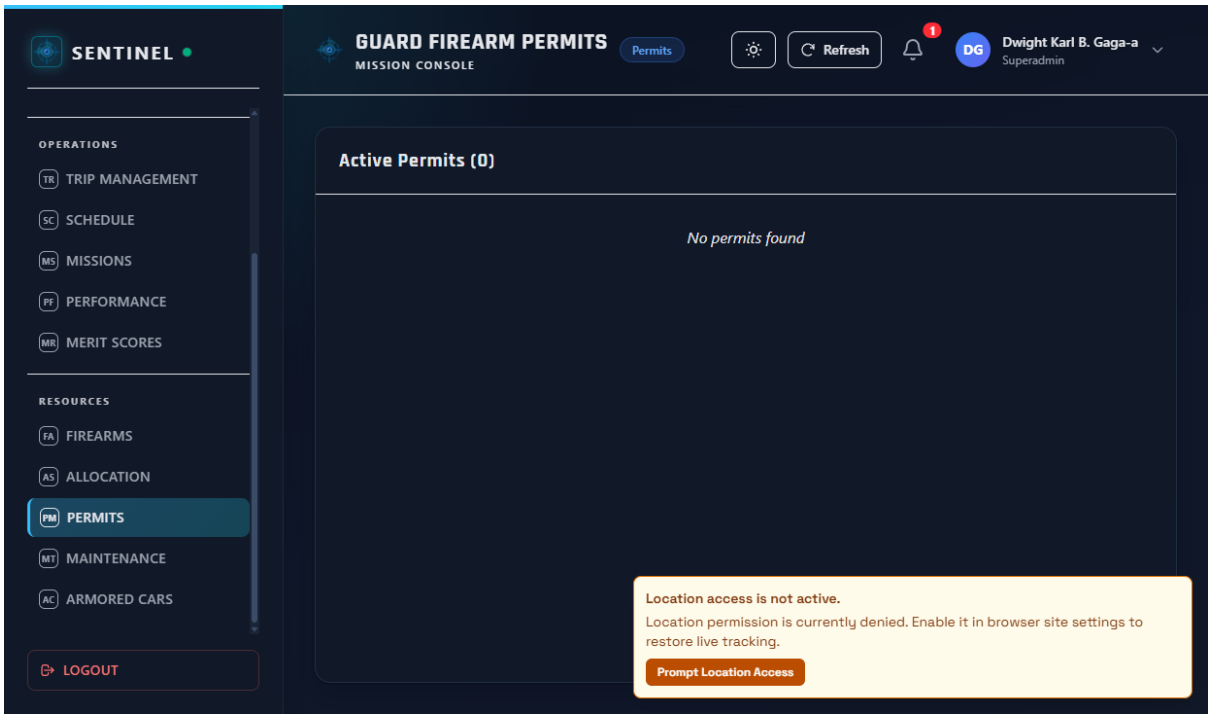

Figure 18. Asset and Compliance Module

- Analytics and Reporting Module

This module will provide operational summaries, trend analysis, reliability indicators, and report generation for management and audit purposes. It will support data-driven decisions across workforce, compliance, and incident management activities.

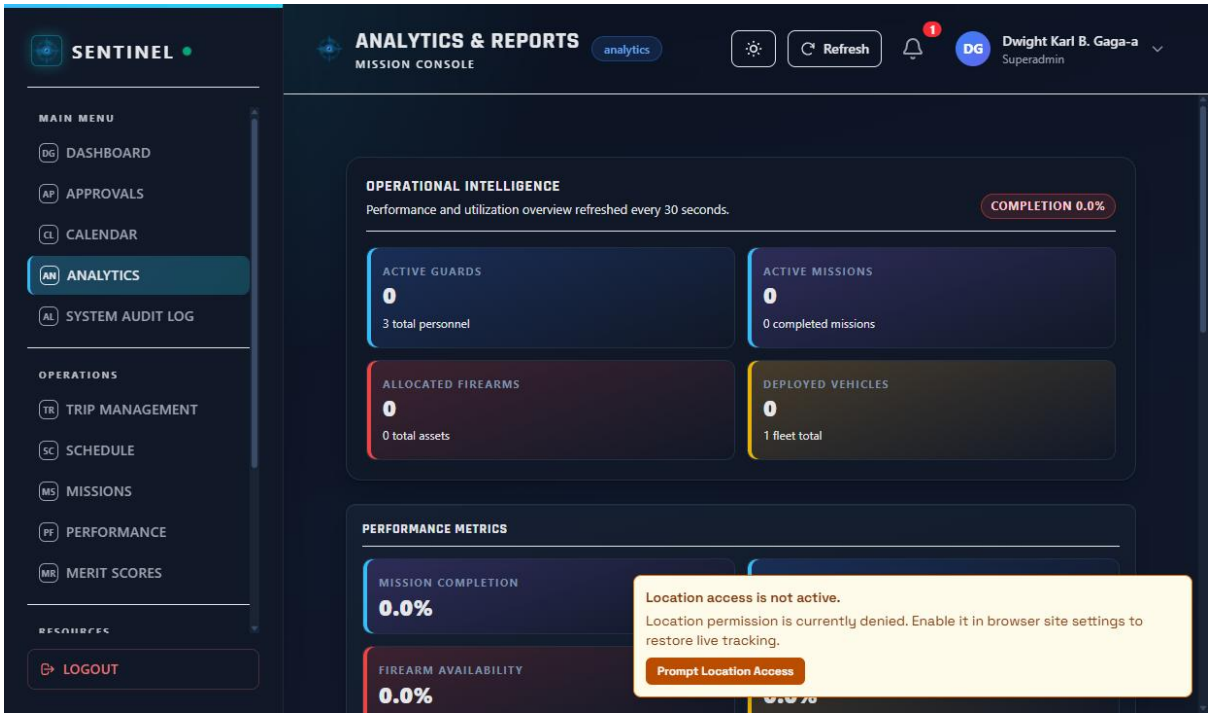

Figure 19. Analytics and Reporting Module

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